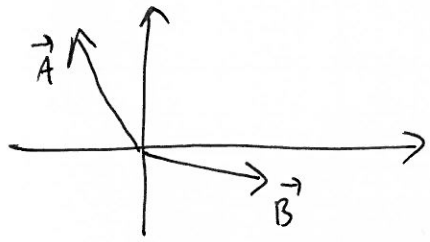
P 1.55 (a) $\vec{A} = -2.00\hat{i} + 6.00\hat{j}$; $\vec{B} = 2.00\hat{i} - 3.00\hat{j}$

$$\vec{A}\cdot\vec{B} = -4.00 - 18.00 = -22.00$$

$$A = \sqrt{2^2+6^2} = \sqrt{40}$$

$$B = \sqrt{2^2+3^2} = \sqrt{13}$$



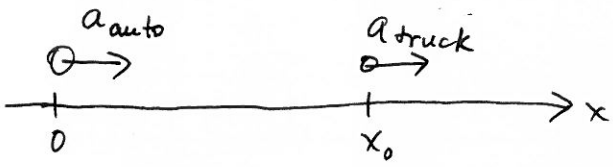
$$\cos\theta = \frac{-22.00}{\sqrt{40}\,\sqrt{13}} = -0.965$$

$$\Rightarrow \theta = 165^\circ$$

(b) $$\cos\theta = \frac{(3)(10)+(5)(6)}{\sqrt{3^2+5^2}\,\sqrt{10^2+6^2}} = 0.882 \Rightarrow \theta = 28^\circ$$

(c) $$\vec{A}\cdot\vec{B} = -(4)(7) + (2)(14) = 0 \Rightarrow \theta = 90^\circ.$$

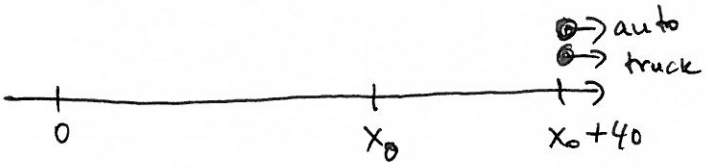
P 2.69 $t=0$:



$a_{auto} = 3.40\ m/s^2$

$a_{truck} = 2.10\ m/s^2$

$v_0 = 0\ m/s$ for both



(a) $$\Delta x_{truck} = 40 = \underset{x_0+40}{\underset{\parallel}{x_{f(truck)}}} - \underset{x_0}{\underset{\parallel}{x_{i(truck)}}}$$

$$\Delta x_{truck} = \cancel{v_0 t} + \frac{1}{2} a_{truck} t^2 \Rightarrow t = \sqrt{\frac{2\cdot 40\,m}{2.10\,m/s^2}} = 6.17\,s$$

(b) Automobile's displacement $$\Delta x_{auto} = \underbrace{(x_0+40)}_{\text{when catches up w/ truck}} - \underbrace{(0)}_{\text{initial}}$$

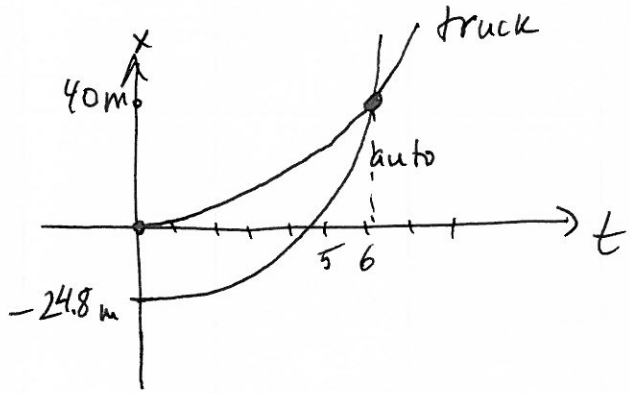
but $\Delta x_{auto} = \frac{1}{2} a_{auto} t^2 = \frac{1}{2}\, 3.40\ m/s^2 \cdot (6.17s)^2 = 64.8\,m$

$\Rightarrow x_0 + 40 = 64.8\ m \Rightarrow x_0 = 24.8\,m$

(c) $v_{auto} = a_{auto}\, t = 3.40 \cdot 6.17 = 21.0\ m/s$

$v_{truck} = a_{truck}\, t = 2.10 \cdot 6.17 = 13.0\ m/s$

(d)



P 2.74 $v_x(t) = \alpha - \beta t^2$, $\alpha = 4.00\ m/s$, $\beta = 2.00\ m/s^3$

$x(0) = 0$

(a) $x(t) - \underset{\overset{\shortparallel}{0}}{x(0)} = \int_0^t d\tau\, v_x(\tau) = \alpha t - \frac{1}{3}\beta t^3$

$a(t) = \frac{dv_x(t)}{dt} = -2\beta t < 0$ for all $t > 0$.

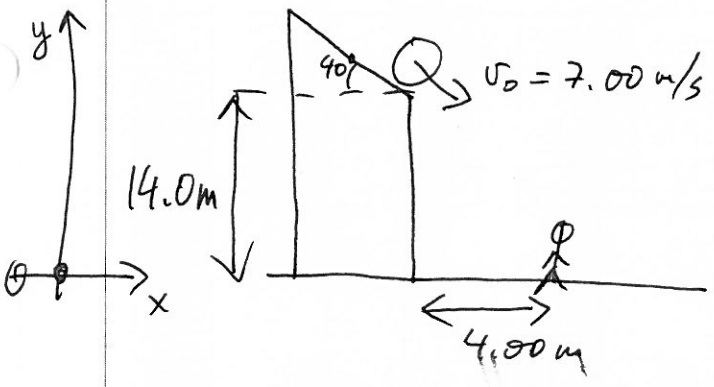
(b) Max of $x(t)$? $\Rightarrow \begin{cases} \frac{dx}{dt} = 0 \Rightarrow v_x(t) = 0 \\ \frac{d^2x}{dt^2} < 0 \Rightarrow a(t) < 0 \end{cases}$

$v_x(t) = 0 = \alpha - \beta t^2 \Rightarrow t = \sqrt{\frac{\alpha}{\beta}} = \sqrt{2}\,s = 1.41\,s$

$a\left(t = \sqrt{\frac{\alpha}{\beta}}\right) = -2\beta\sqrt{\frac{\alpha}{\beta}} = -2\sqrt{\alpha\beta} < 0$

$x\left(t = \sqrt{\frac{\alpha}{\beta}}\right) = \left(\alpha - \frac{\beta}{3}\frac{\alpha}{\beta}\right)\sqrt{\frac{\alpha}{\beta}} = \frac{2}{3}\alpha\sqrt{\frac{\alpha}{\beta}} = 3.77\ m$

P3.60



(a) $y_f = 0 = y_i - v_0 \sin 40° t - \frac{1}{2} g t^2$

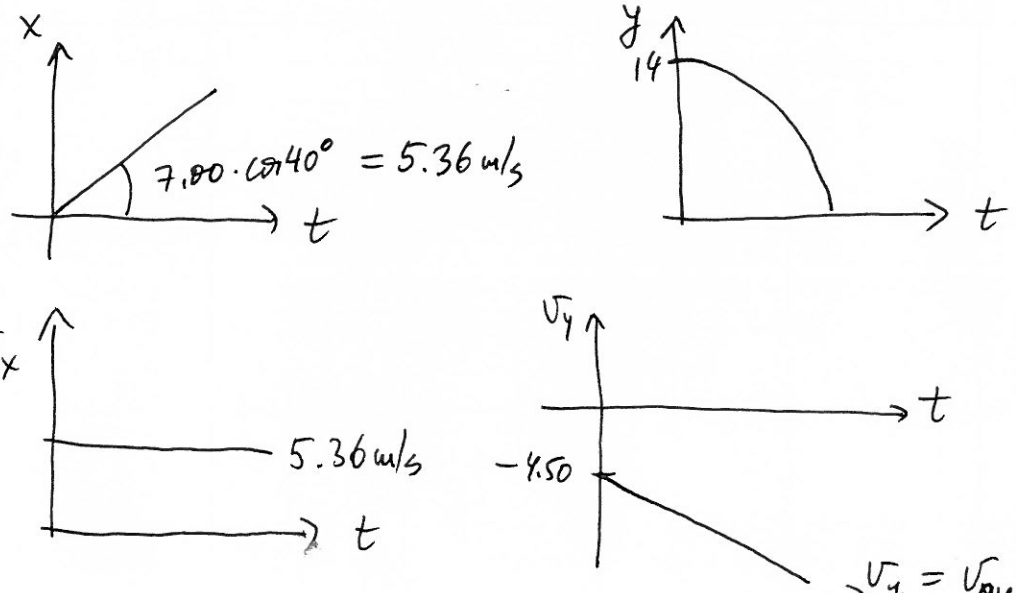
$y_i = 14.0\text{m}$

$$4.90 t^2 + 7.00 \sin 40° \, t - 14.0 = 0$$

$$t = \frac{-7 \cdot \sin 40° \pm \sqrt{(7 \sin 40°)^2 - 4(4.90)(-14.0)}}{9.80} = \frac{-4.50 \pm 17.2}{9.80}$$

choose positive root: $t = 1.29\text{s}$

(b)



$v_y = v_{0y} - gt$
$= -4.50 - 9.80t$

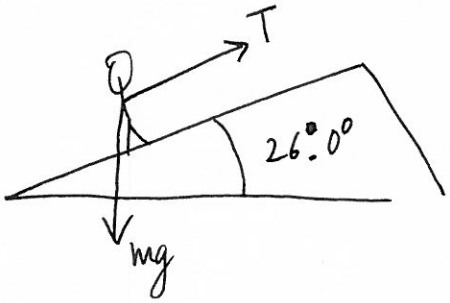
(c) Snowball travels $\Delta x = 4.0\text{m}$ in $\Delta t = \frac{4.0\text{m}}{5.36\text{m/s}} = 0.746\text{s}$

Its vertical displacement is

$$\Delta y = -v_0 \sin 40° \Delta t - \frac{1}{2} g (\Delta t)^2 = -6.08 \text{ m}$$

$\Rightarrow$ snowball is at $y = 14.0 - 6.08\text{m} = 7.9\text{m} > 1.9\text{m}$ above the man.

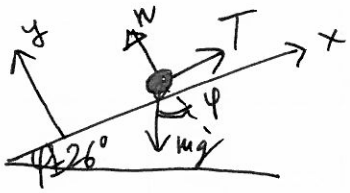
P4.32



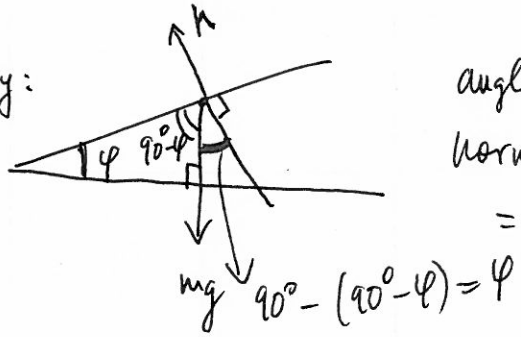
no friction

$\vec{v} = \text{const} \Rightarrow \underline{\vec{a} = 0}$!

free-body diagram:



geometry:



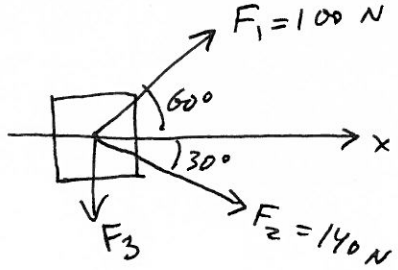
angle between normal $\vec{n}$ and $m\vec{g}$ = = incline angle = φ

along y: $n - mg\cos\varphi = 0$

along x: $T - mg\sin\varphi = 0$

$T = mg\sin 26^\circ = 279\ N.$

P4.37



(a) In order to move along x,

$\sum F_y = 0 \Rightarrow F_1 \sin 60^\circ - F_2 \sin 30^\circ + F_{3y} = 0$

$F_{3y} = 140\sin 30^\circ - 100\sin 60^\circ =$
$= -16.6\ N$

Smallest $\vec{F}_3$: $F_{3x} = 0$, $F_{3y} = -16.6\ N$

(b) $\sum F_x = F_1\cos 60^\circ + F_2\cos 30^\circ = m(2.0\ m/s^2)$

$m = 85.6\ kg \Rightarrow W = mg = 839\ N.$